

Do CRRT Machine Signals add prognostic value?

A multimodal machine learning approach to ICU Mortality Prediction

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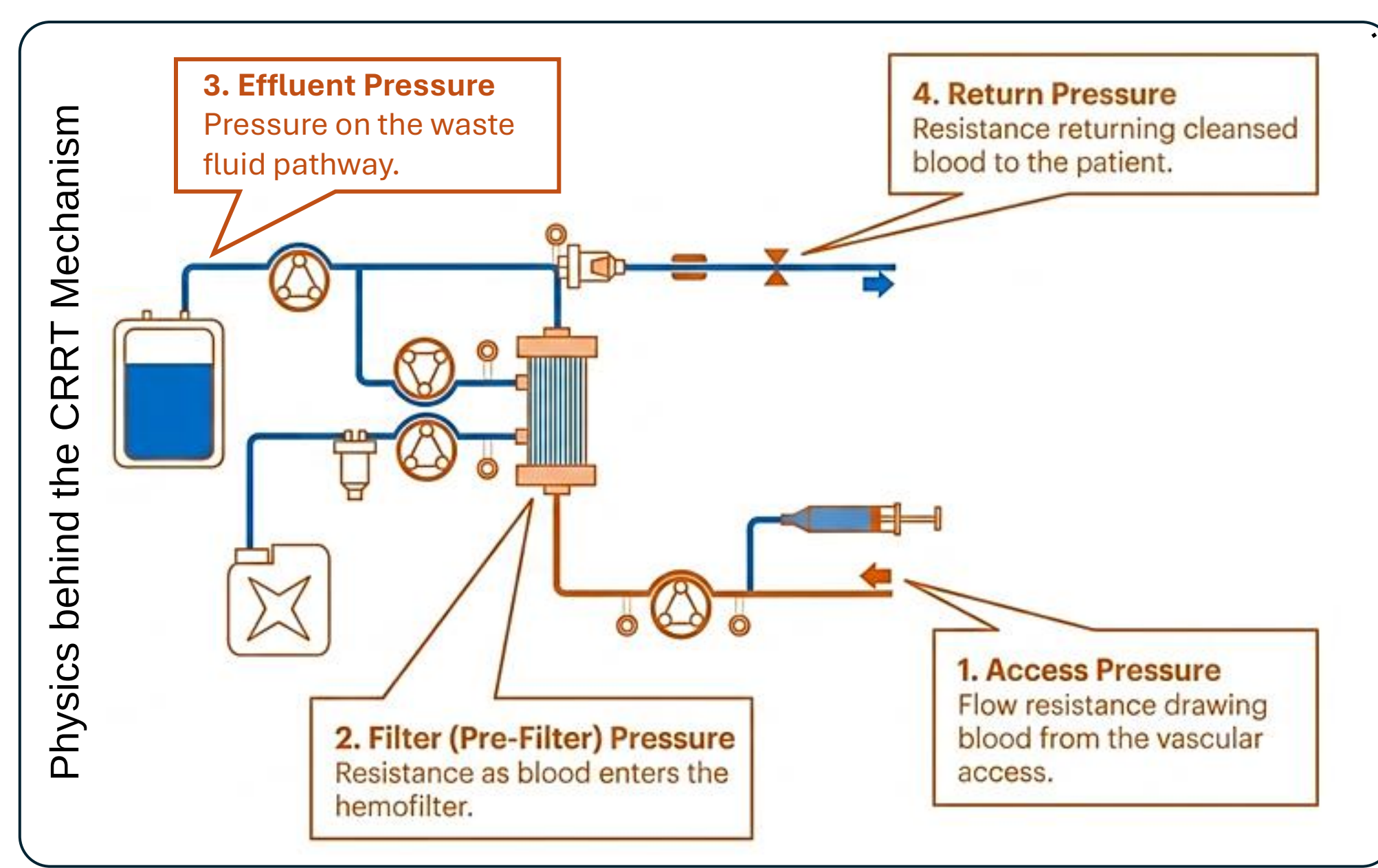
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BACKGROUND AND PROBLEM STATEMENT

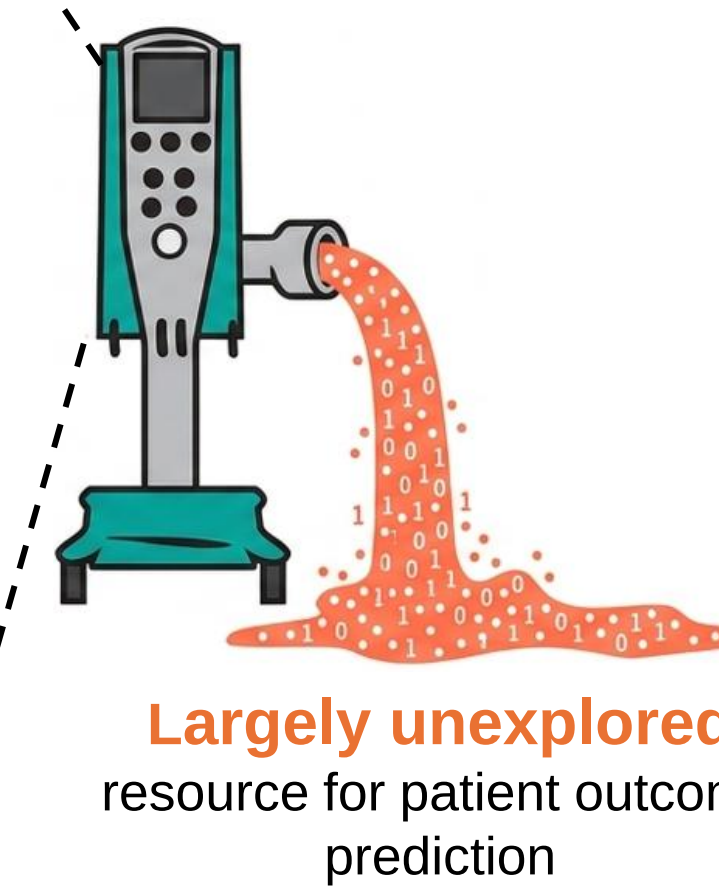


~50% Mortality Rate in ICU-AKI

AKI requiring renal replacement therapy (RRT) affects 5-15% of ICU admissions and remains high-risk clinical challenge.



CRRT machine generates a high-frequency operational data **every minute**, capturing every pressure, flow and alarm. But, it is **mostly discarded** by EHR-based prediction models.



Largely unexplored resource for patient outcome prediction

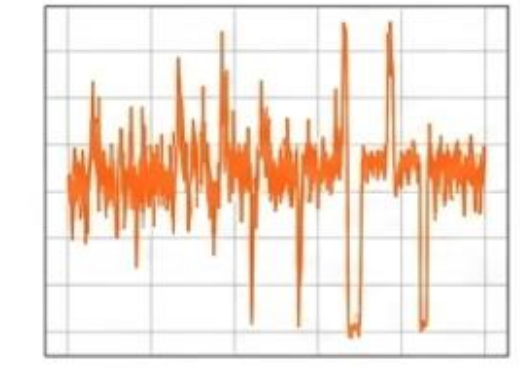
The Challenge

Overwhelming Volume

Tens of thousands of points per treatment; impossible for manual human review.

Noise & Artifacts

Pressure streams are highly corrupted by patient movement, nursing interventions, and line manipulation (spikes and downtime).



No Interpretation Framework

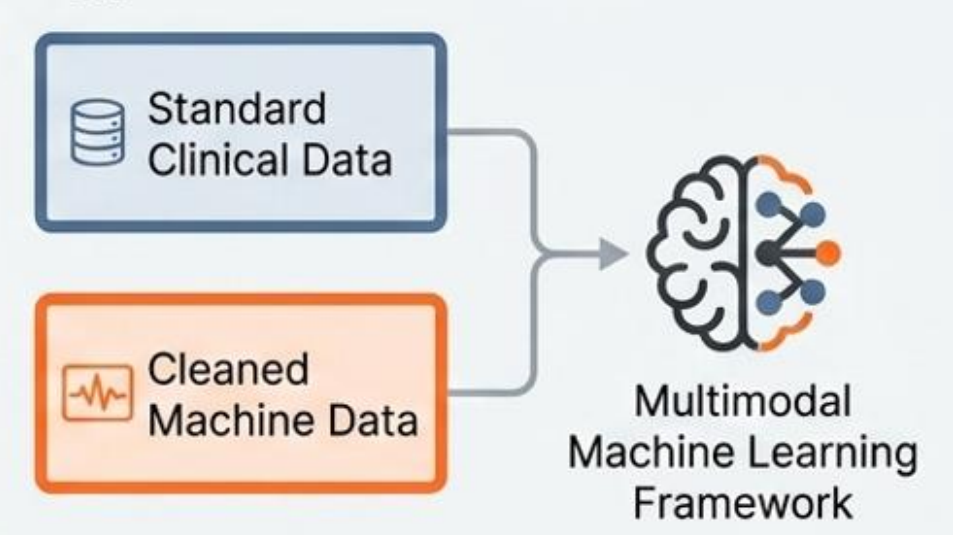
No accepted clinical rules exist based on these continuous pressure trajectories.

The Objective

Goal:

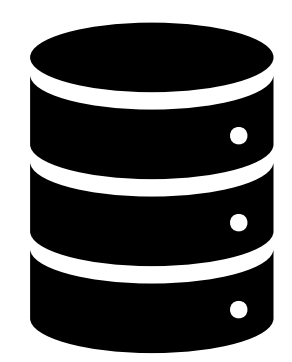
Evaluate whether adding cleaned CRRT pressure features to clinical data improves rolling day-wise mortality prediction over clinical data alone.

Approach:



METHODS & STUDY COHORTS

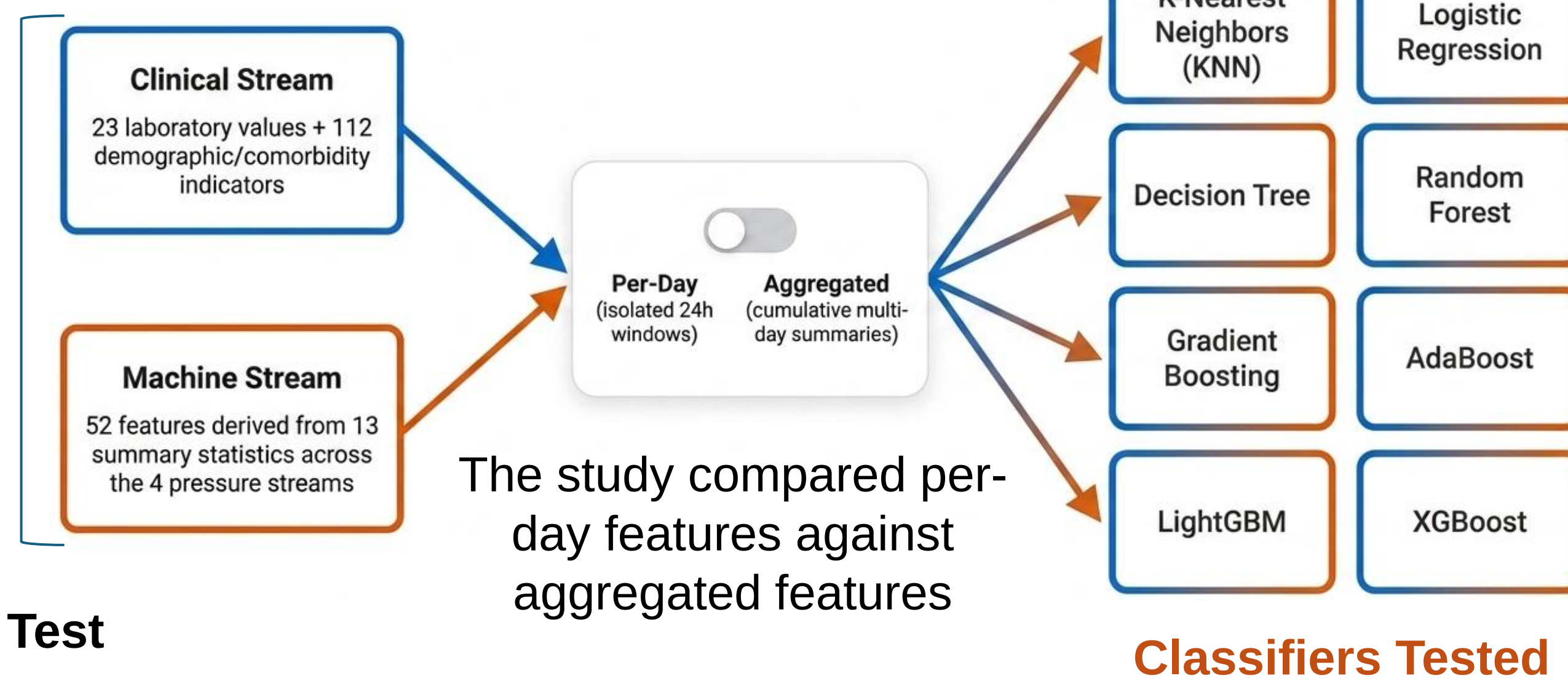
CRRTnet Registry



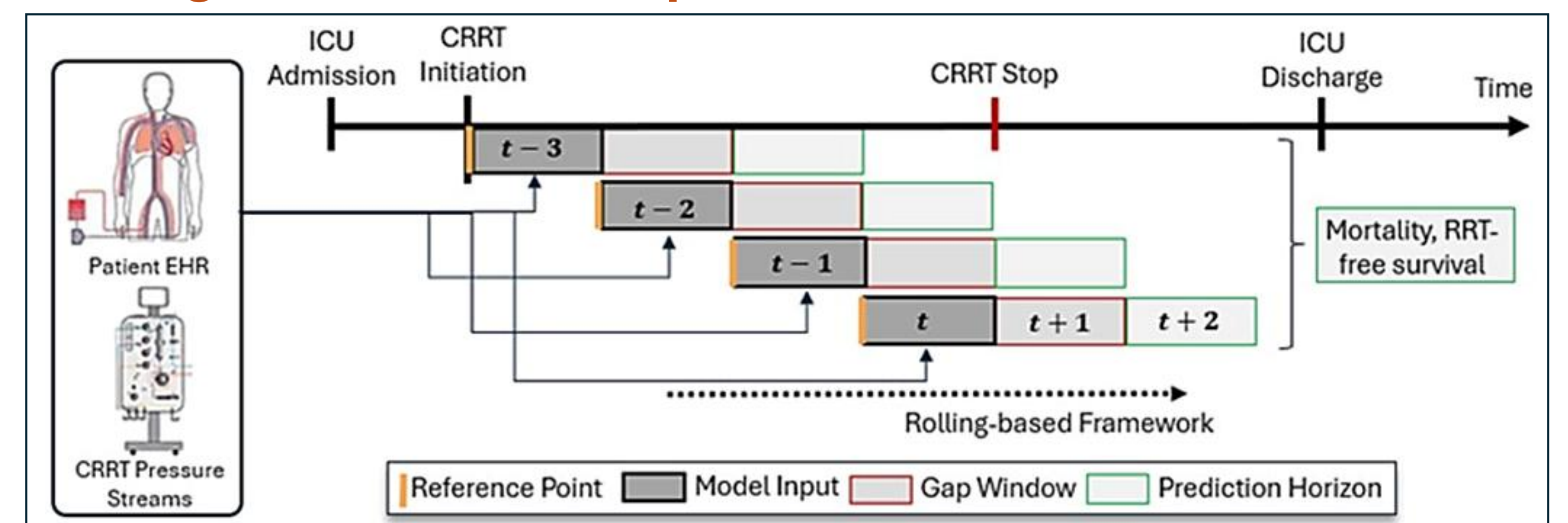
976 Unique Patients,

5,193 Treatment-Days

70% Train 15% Val 15% Test



Rolling-based outcome prediction framework



Prediction were made using **1-day** windows, including a 'gap window' to prevent data leakage and ensure the model predicts future outcome

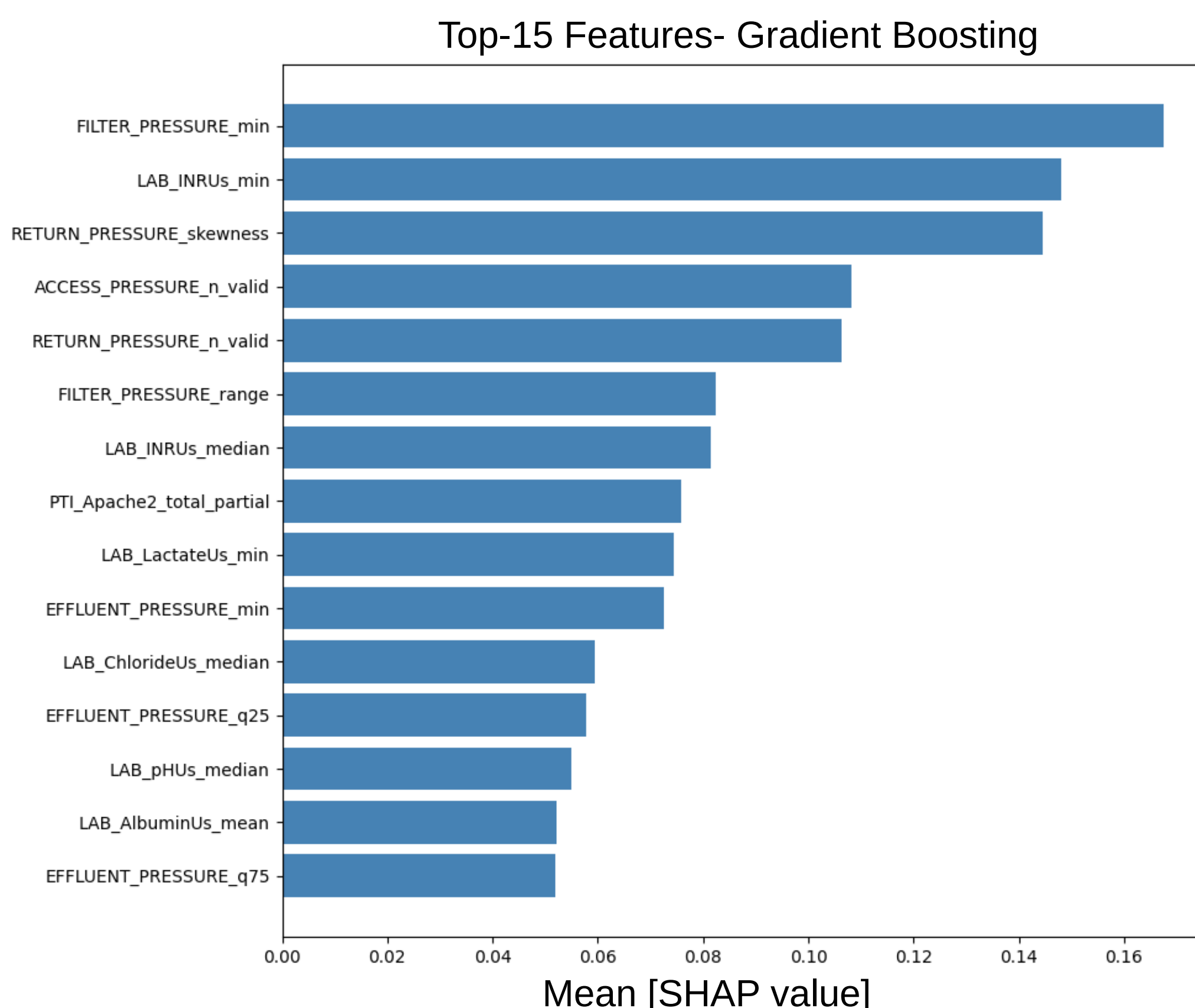
RESULT ANALYSIS

Performance Comparison Table

Feature Source	Feature Strategy	Best Model	AUROC	F1
Clinical Only	Per-Day	RF	0.730	0.702
Clinical Only	Aggregated	RF	0.776	0.718
Machine Only	Per-Day	LR	0.686	0.671
Machine Only	Aggregated	AdaBoost	0.687	0.658
Machine + Clinical	Per-Day	RF	0.753	0.696
Machine + Clinical	Aggregated	GB	0.801	0.759

Note: Aggregation improved the 1-Day combined pipeline

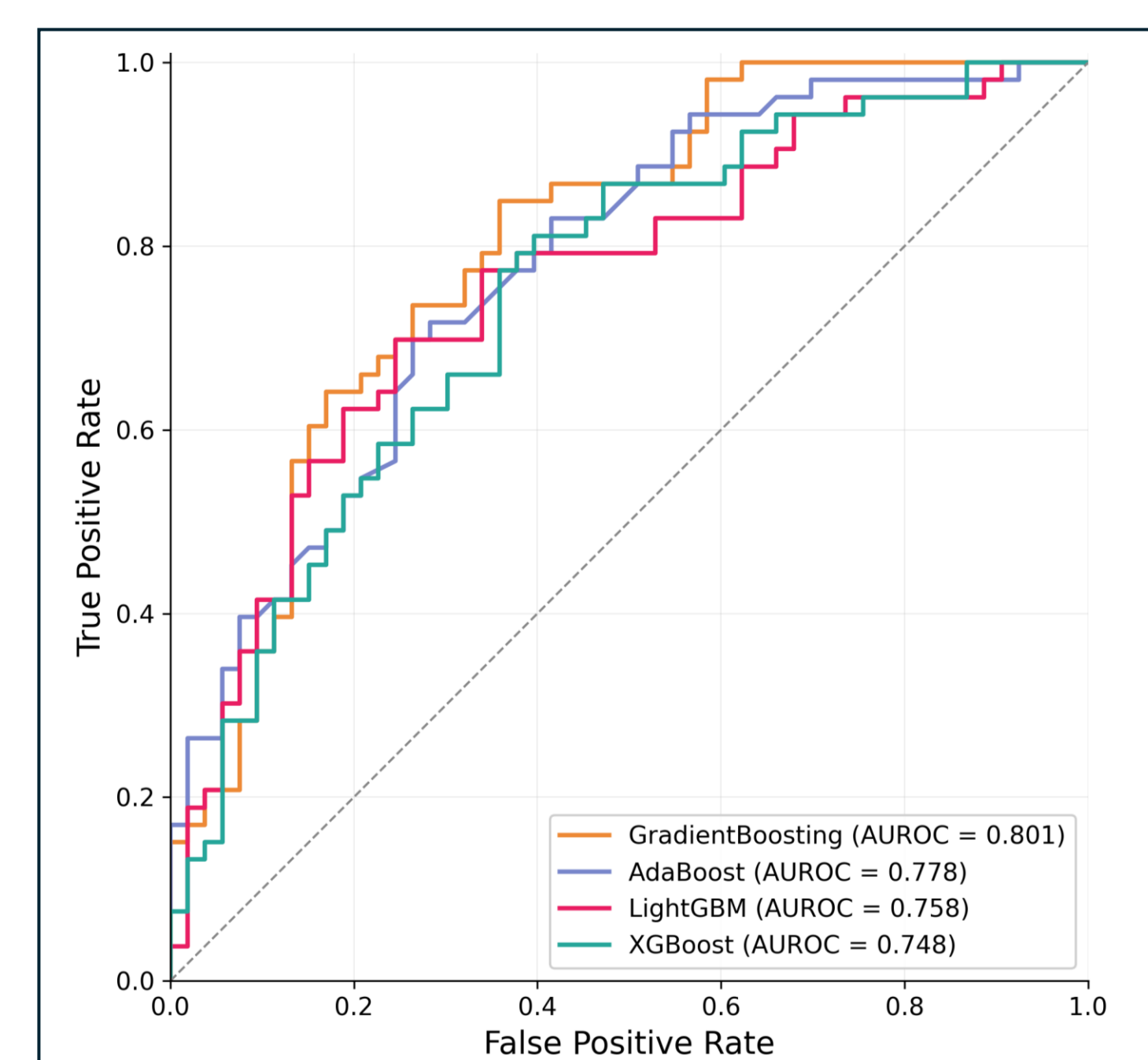
SHAP Summary Plot (Feature Importance)



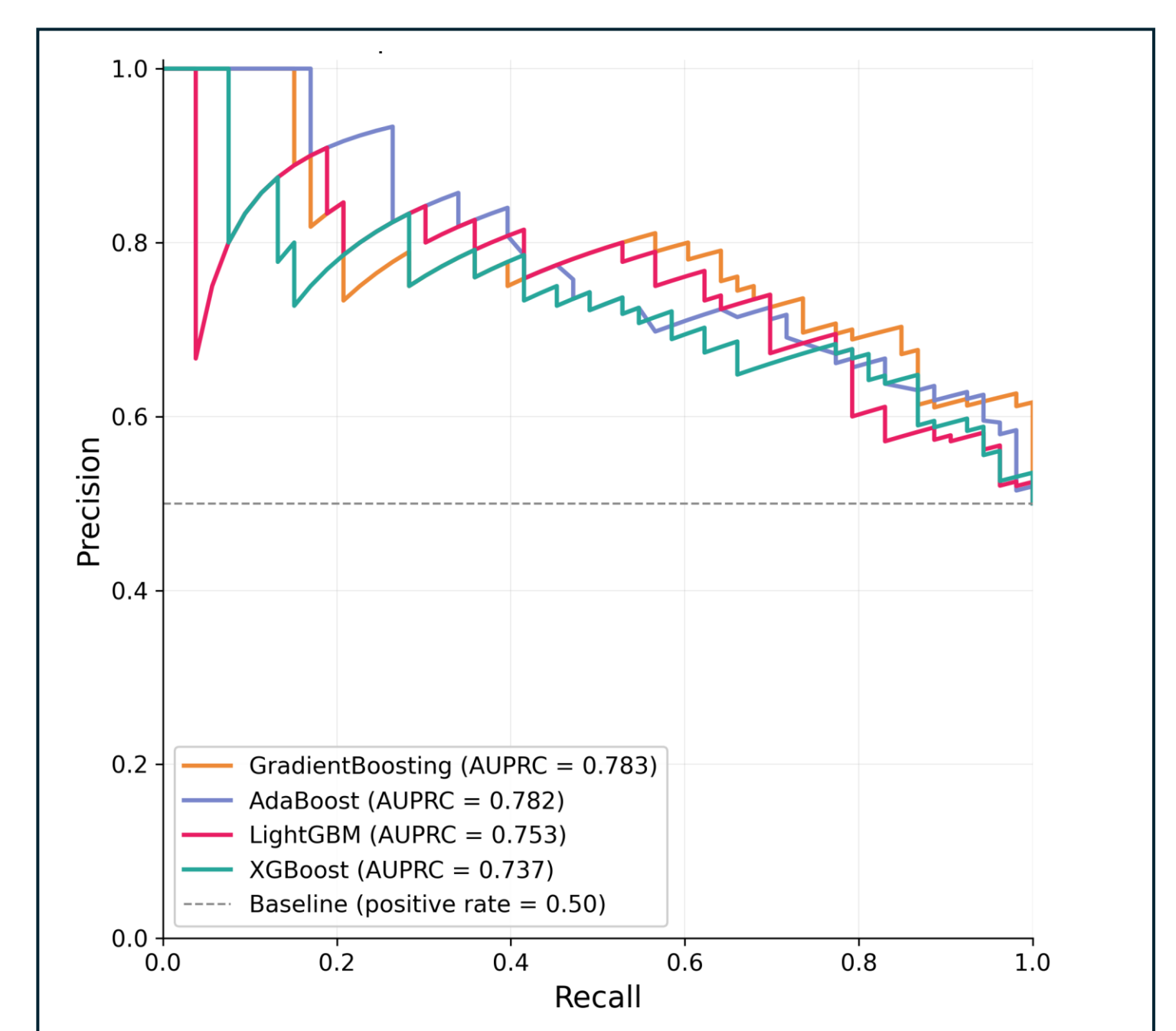
1-Day



ROC and Precision-Recall Curves (Top-4 Models. 1-Day Combined)

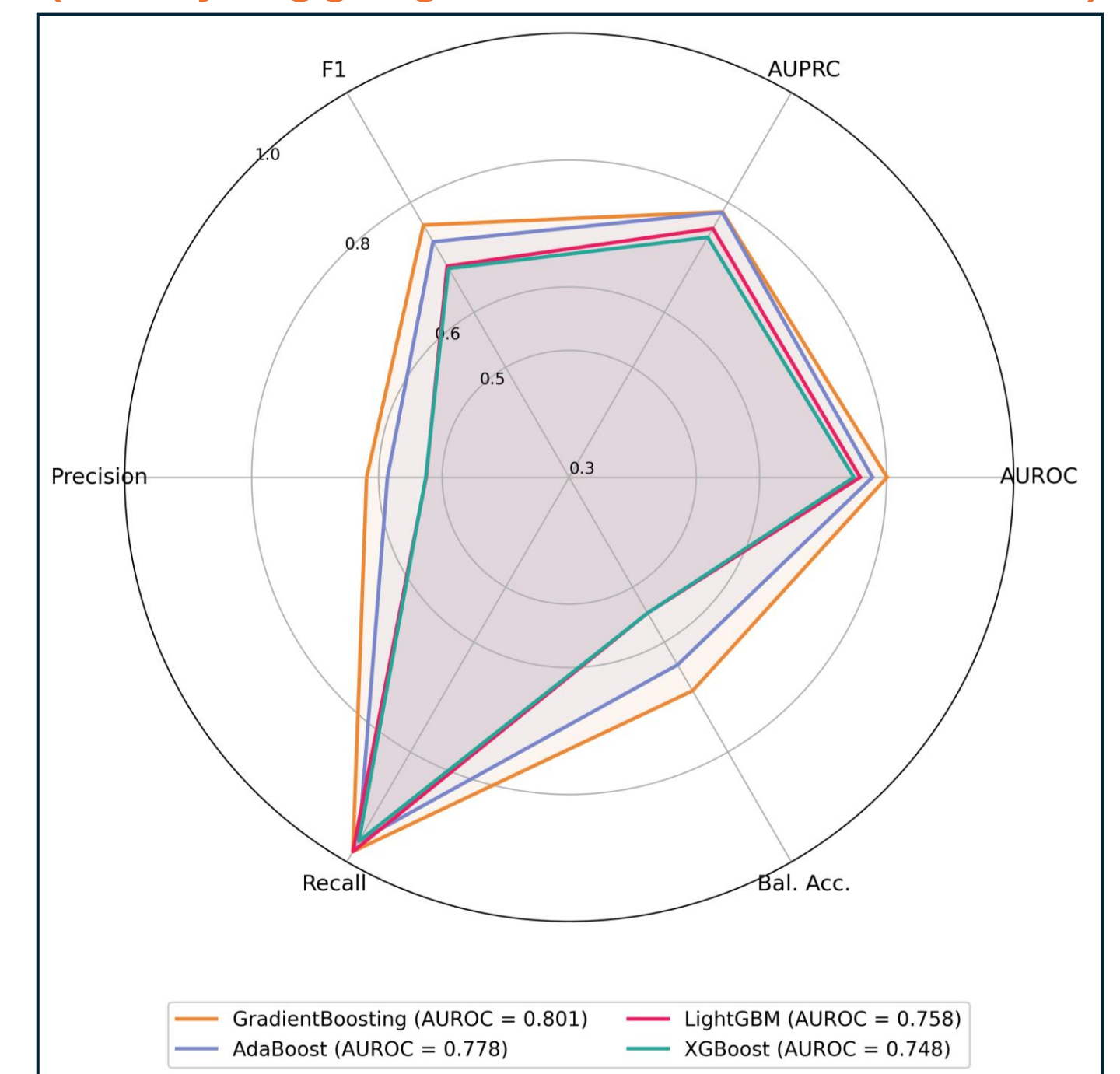


ROC Curve



Precision-Recall Curve

Multi-metric Radar Plot (1-Day Aggregated Machine + Clinical)



Machine vs. Clinical Contribution

While clinical features remain primary drivers, machine data provides a disproportionate signal, occupying 53% of top-15 predictive positions despite being only 28% of the input space

CONCLUSION

- CRRT machine hold critical, uncaptured prognostic information. Integrating discarded CRRT machine data with clinical variables significantly improves ICU mortality prediction (AUROC > 0.80)
- Cumulatively summarizing multi-day historical data provides a decisive advantage over isolated 24-hour snapshots

FUTURE WORK

- An AI that watches CRRT machine signals minute by minute and learns on its own which moments matter most for patient outcomes, making real-time bedside alerts.
- A first knowledge-graph-guided multimodal CRRT framework that aligns and integrates machine signals, EHR variables, and clinical notes, enabling bedside CDS to reason across modalities the way clinicians do.

ACKNOWLEDGEMENT

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